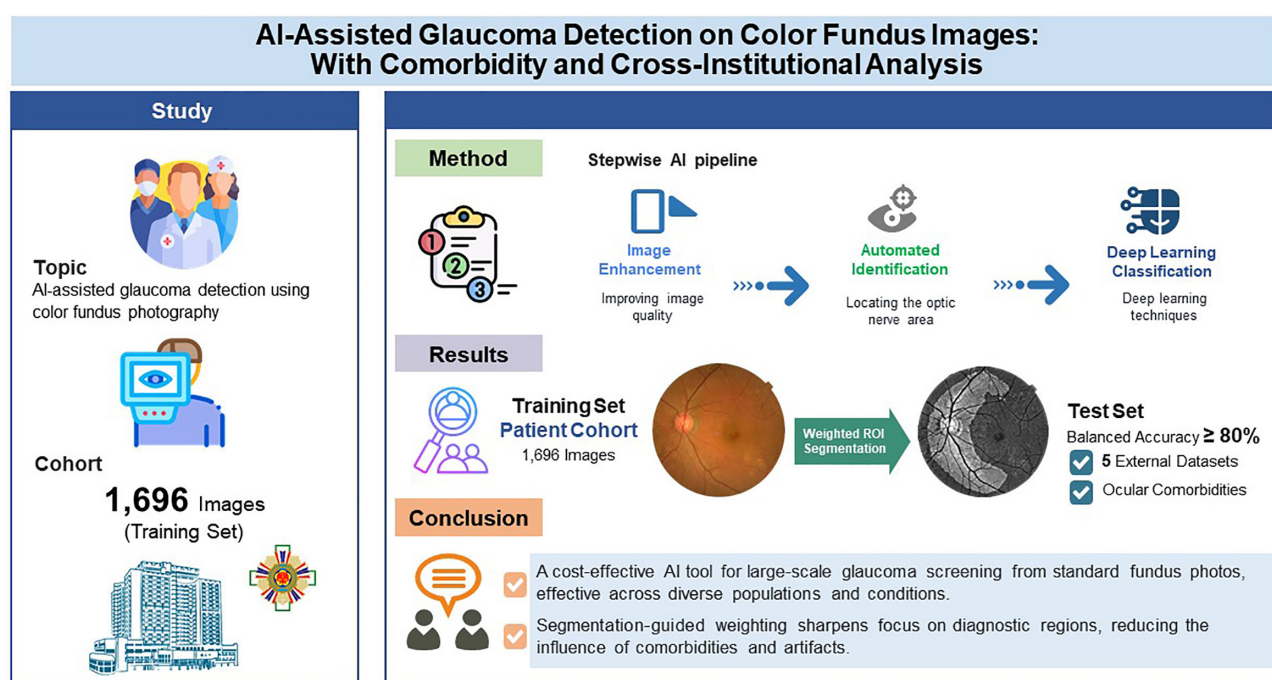


Artificial intelligence-assisted glaucoma detection on color fundus images: with comorbidity and cross-institutional analysis

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Abstract

Background: Glaucoma is a major cause of irreversible blindness, and early detection is essential to prevent vision loss. Color fundus photography is a simple, low-cost, and noninvasive eye screening method, but diagnosis through this method can be difficult in patients with additional retinal diseases. Although artificial intelligence (AI) can address this difficulty, its effectiveness may vary between hospitals. In this study, an AI glaucoma detection system was developed and tested for reliability across different populations and clinical settings.

Methods: A stepwise AI pipeline was designed that combined image enhancement, automated identification of the optic nerve area, and deep learning-based classification. The system was trained on 1696 images from Taipei Veterans General Hospital and tested on five cross-regional external datasets. The system was also evaluated on a separate internal set of 151 images representing comorbid eye diseases.

Results: The AI system achieved a balanced accuracy of at least 80% on all external datasets. For images with other eye diseases, it achieved an area under the curve of 0.93 and a balanced accuracy of 80.9%. Its performance remained consistent regardless of differences in patient ethnicity, camera types, and image quality.

Conclusion: The proposed AI system can detect glaucoma on standard color fundus photographs with high accuracy across clinical environments and in the presence of comorbid eye diseases. The system may be a practical and affordable tool for large-scale glaucoma screening, particularly in institutions with limited resources.

Keywords: Color fundus; Comorbidities; Cross-institutional; Deep learning; Glaucoma

Lay Summary: Glaucoma is a major cause of vision loss worldwide, and early detection is critical to prevent blindness. However, current screening methods often require expensive equipment and expert interpretation, rendering them difficult to implement in many clinics. In this study, we developed a low-cost, artificial intelligence-powered system that can detect signs of glaucoma on commonly taken eye images (color fundus photos). Whereas other methods may be less effective when differences between hospitals, patients, or image quality are present, our model is designed to work reliably across institutions, devices, and patients, including those with comorbid eye diseases. We trained the model to focus only on the most crucial parts of the eye, mimicking how doctors examine the optic nerve. The system was tested on multiple international datasets and achieved high accuracy. This approach has strong potential for use in real-world healthcare settings, particularly in institutions with limited medical resources, enabling wider and earlier glaucoma screening.

1. INTRODUCTION

Glaucoma is a leading cause of irreversible blindness. Early detection of this condition is challenging, and accurate diagnosis typically requires multiple evaluations by experienced specialists using various medical imaging tools. Color fundus imaging is widely used to identify suspicious optic discs because it is easily accessible, affordable, and noninvasive. Numerous studies have demonstrated the diagnostic potential of glaucoma detection algorithms based on color fundus imaging.^{1,2} The following text outlines the progression of medical imaging techniques for glaucoma detection, spanning the initial development of imaging tools and processing algorithms to the application of machine and deep learning techniques for detection. These techniques have considerably improved the efficiency of glaucoma detection on color fundus images.

Red-free light fundus photography was first introduced in 1972. It substantially improved glaucoma detection by increasing the visibility of superficial retinal reflections, particularly the stripe reflections of nerve fiber bundles.^{3–6} In 2008, an image processing algorithm was developed to measure the optic disc and optic cup areas, enabling providers to calculate the cup-to-disc ratio (CDR) and facilitating large-scale automatic screening for glaucoma.⁷ In 2016, the application of machine learning and image processing to glaucoma detection marked a further breakthrough, substantially improving the efficiency of automated CDR calculations.⁸ In 2017, deep learning technology was introduced, enabling the application of convolutional neural networks (CNNs) as feature engineering modules. After image features are extracted, these modules match classifiers with machine learning algorithms or fully connected layers to train glaucoma detection models.^{9,10} Deep learning has also been used to automatically segment and calculate the CDR, demonstrating its potential for cross-dataset applications.^{11,12} In 2022, the use of diverse datasets to train models enhanced their versatility,¹³ and in 2023, the integration of multiple deep learning models designed for image quality detection and glaucoma disease segmentation and detection further enhanced glaucoma detection.¹⁴

Scholars in this area of research have primarily focused on developing algorithms for single datasets. However, models must be generalizable and cost-effective for artificial intelligence (AI)-assisted glaucoma detection to be widely applicable. Accordingly, in this study, we created a cost-effective generalized glaucoma detection model using color fundus images. Fundus images are of high resolution and contain vivid colors, and therefore, strategies are required to address variations in image characteristics arising from differences in camera equipment and patient characteristics, such as race and comorbidities. In addition, maintaining consistent performance across diverse datasets and images reflecting various comorbidities can be challenging. We therefore integrated several image processing methods and semantic segmentation models to develop a robust feature extraction process. Weighted images were used to fine-tune the pretrained classification model. Diagnostic algorithms based on fundus images, which are readily available in hospitals and clinics, have considerable potential for real-world application. We created such an algorithm through a multistep process:

1. Like red-free images, color fundus images retain only the green light wavelength, enhancing image features. We thus applied image processing algorithms to the green channel to improve feature contrast and sharpness.
2. Key diagnostic areas were identified through semantic segmentation, which avoided the influence of noise and the confounding effects of comorbidities.
3. Domain knowledge was used to assign increased weight to areas critical for glaucoma detection, which further reduced the effects of noise and comorbidities.

We validated our method in multiple external datasets, demonstrating a balanced accuracy of 80%, which is comparable to that of commercial optical coherence tomography (OCT) images.¹⁵ Furthermore, our use of color fundus images offers considerable cost advantages, underscoring our method's promise for use in large-scale glaucoma screening.

2. METHODS

Fig. 1 illustrates the study workflow. We leveraged image pre-processing techniques to extract the green channels of color fundus images and enhance the contrast and sharpness of image features. After preprocessing, we used the DeepLabV3+ architecture¹⁶ to construct a semantic segmentation model capable of selecting critical areas for glaucoma detection. The DeepLabV3+ model, which uses transfer learning, was trained on images from Taipei Veterans General Hospital (TVGH). We then applied the segmentation results from DeepLabV3+ to weight the original image regions. These weighted images were used to train the EfficientNetV2-M¹⁷ transfer learning model, creating our glaucoma detection classifier, referred to hereafter as the TVGH model.

The diagnostic accuracy of the TVGH model was evaluated according to test images from TVGH and five external datasets, including four open datasets (DRISHTI-GS1,¹⁸ VEIRC,^{19,20} Refuge,²¹ and LES-AV²²) and one private dataset (Cheng Hsin General Hospital [CHGH]). Our comprehensive glaucoma detection model integrates image feature enhancement and automatic segmentation of critical areas for disease detection, which enhances its predictive performance on external datasets.

2.1. Data collection

The TVGH dataset included 1075 nonglaucomatous eyes from the TVGH Ophthalmic Imaging Database and 621 fundus photographs of eyes with glaucoma in the JPEG format. Glaucoma

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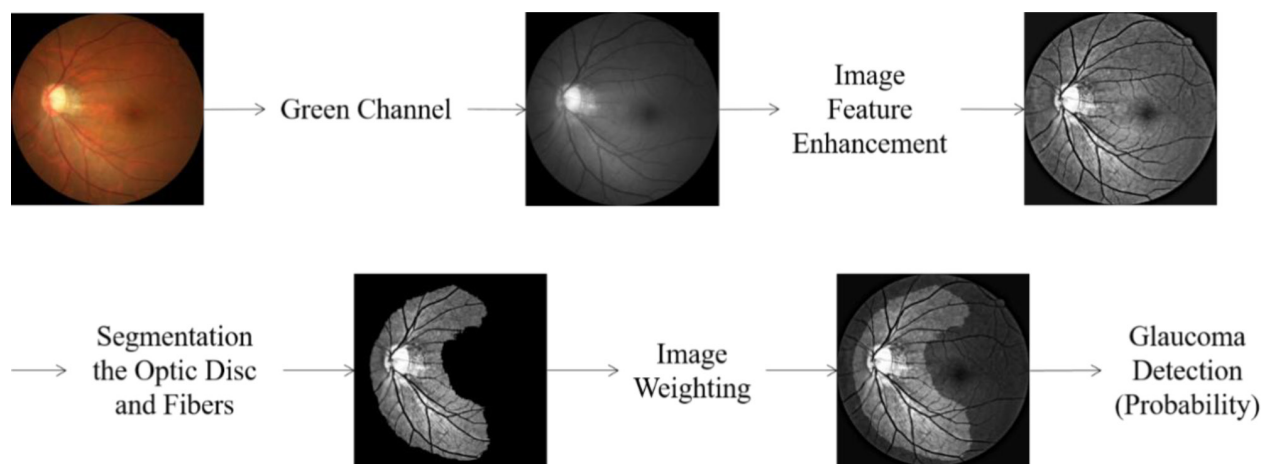


Fig. 1 Overview of proposed glaucoma detection workflow, encompassing preprocessing, segmentation, and classification stages.

was diagnosed on the basis of characteristic changes in the optic nerve head (ONH) and associated reproducible visual field defects. Glaucomatous ONH changes were defined as an elevated CDR and generalized neuroretinal rim thinning or localized notches in the rim. All patients with glaucoma underwent visual field analysis with the 24-2 Swedish Interactive Threshold Algorithm standard on a Humphrey Field Analyser 750i (version 4.2, Zeiss-Humphrey Instruments, Dublin, CA) and retinal nerve fiber layer (RNFL) scanning with a Cirrus high-definition OCT device (model 4000; Carl Zeiss Meditec, Dublin, CA). The nonglaucomatous control eyes had intraocular pressure of <21 mmHg and an ONH of normal appearance without RNFL defects. Images with severe artifacts, excessive blurring, or loss of visibility in the ONH or RNFL were excluded from the training set. Additionally, overexposed optic discs were excluded through careful visual assessment by glaucoma specialists to ensure adequate image quality for model training. Only images in which the main diagnostic features remained visible after preprocessing were included in the training set.

Table 1 lists the imaging characteristics of the TVGH and external datasets. Images were classified on the basis of the majority opinion of experts for the DRISHTI-GS1, Refuge, VEIRC, and LES-AV datasets and through expert consensus after a review of detailed clinical information for the CHGH and TVGH datasets. The TVGH dataset images were divided into training, validation, and test sets in a 64%:16%:20% sampling ratio.

2.2. Image preprocessing

The five-step image preprocessing procedure is illustrated in Fig. 2. First, color fundus images were converted into binary images, and the area outside the retina was replaced with zero padding. This step ensured that the background signal did not interfere with subsequent CNN modeling. Second, the image was cropped to square dimensions along the retinal boundaries. Nonsquare cropped images were padded with zeros to create square dimensions. This step ensured that the input images had the same layout and that the image features were not distorted by resizing when added to the

Table 1
Characteristics of the datasets used in this study

Dataset	Numbers (glaucoma)	Image size (pixels)	Field of view (center)	Camera	Origin
TVGH	1696 (621)	3888 × 2552	30° (macula)	Canon CX-1, CR-2, DGi	Taiwan
CHGH	158 (78)	3216 × 2136	30° (macula)	Topcon TRC-NW8F	Taiwan
DRISHTI-GS1	101 (70)	2047 × 1760	30° (disc)	Zeiss Visucam NM/FA	India
VEIRC	257 (32)	2544 × 1696	25° (macula)	Welch Allyn Pan	India
Refuge	400 (40)	2124 × 2056	(macula)	Zeiss Visucam 500	China
LES-AV	22 (11)	1620 × 1444	30° (disc)	Visucam ProNm	Austria

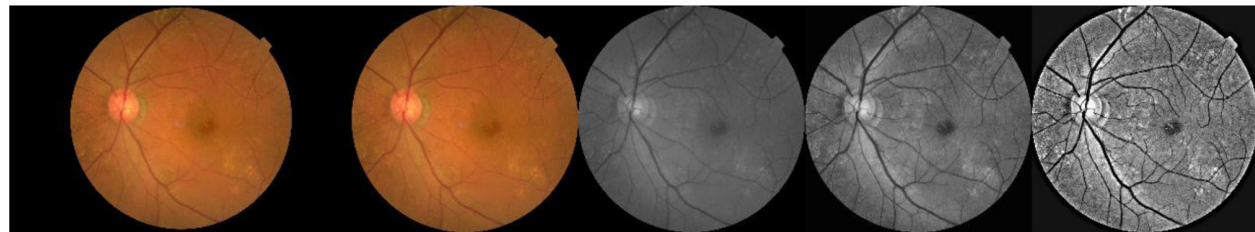


Fig. 2 Five-step image preprocessing pipeline: (1) zero padding outside retinal boundary, (2) cropping and squaring of retinal region, (3) green channel extraction, (4) contrast enhancement with CLAHE, and (5) sharpening through unsharp masking based on Gaussian blur. CLAHE = contrast-limited adaptive histogram equalization.

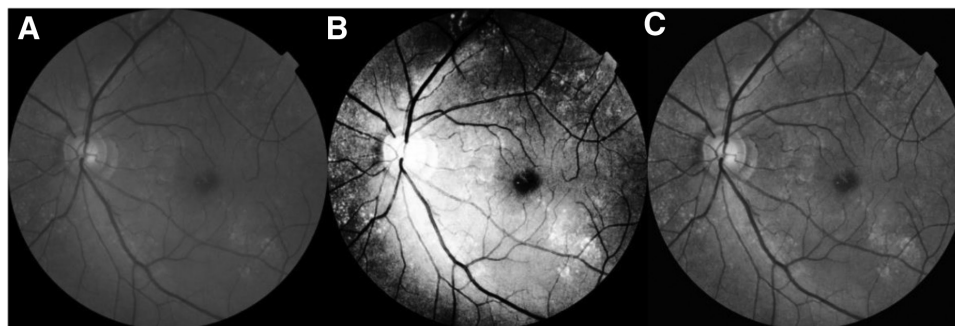


Fig. 3 Comparison of green channel images: (A) original, (B) histogram equalized, and (C) CLAHE-enhanced. CLAHE = contrast-limited adaptive histogram equalization.

CNN model. Third, the green channel was retained in the cropped images, which enabled observation of disease characteristics such as RNFL striation and changes in the vessel contour within the optic disc. Fourth, the contrast-limited adaptive histogram equalization (CLAHE) algorithm was applied for image contrast enhancement.^{23,24} Finally, image sharpness was increased through an unsharp masking process based on Gaussian blur.

2.2.1. Image contrast enhancement with CLAHE

The CLAHE technique is widely used to enhance image features by increasing contrast. This method improves upon histogram equalization by both increasing image contrast and suppressing noise amplification through the application of a brightness threshold. Fig. 3 illustrates the differences between the green channel, the green channel with histogram equalization, and the green channel with CLAHE. Fig. 4 displays the corresponding histogram differences for Fig. 3.

2.2.2. Unsharp masking for enhanced image sharpness

Unsharp masking is a technique that increases image sharpness by emphasizing the image's edges. This method is used to

generate a sharpened image by subtracting a blurred version from the original. Fig. 5 demonstrates differences in the characteristic channel image after the application of both CLAHE and unsharp masking for sharpening.

2.3. Image segmentation and weighting with DeepLabV3+ architecture

Glaucoma specialists determine the presence of glaucoma by examining the ONH and RNFL in color fundus images. Because CNN models are sensitive to subtle changes in image features, numerous characteristics of fundus photographs from different datasets can influence their classification. These characteristics include ethnic variations, differences in fundus cameras and operators, and interference from other retinal conditions, such as retinal hemorrhages, exudates, or other optic neuropathies.

We manually annotated the regions of interest (ROI) for segmentation, focusing on the ONH and surrounding RNFL. The segmented area was extended radially according to the disc radius to encompass major vessel trunks representing the RNFL double-hump region.²⁵ Macular regions were excluded to avoid the confounding effects of unrelated retinal pathologies, as illustrated in Fig. 6.

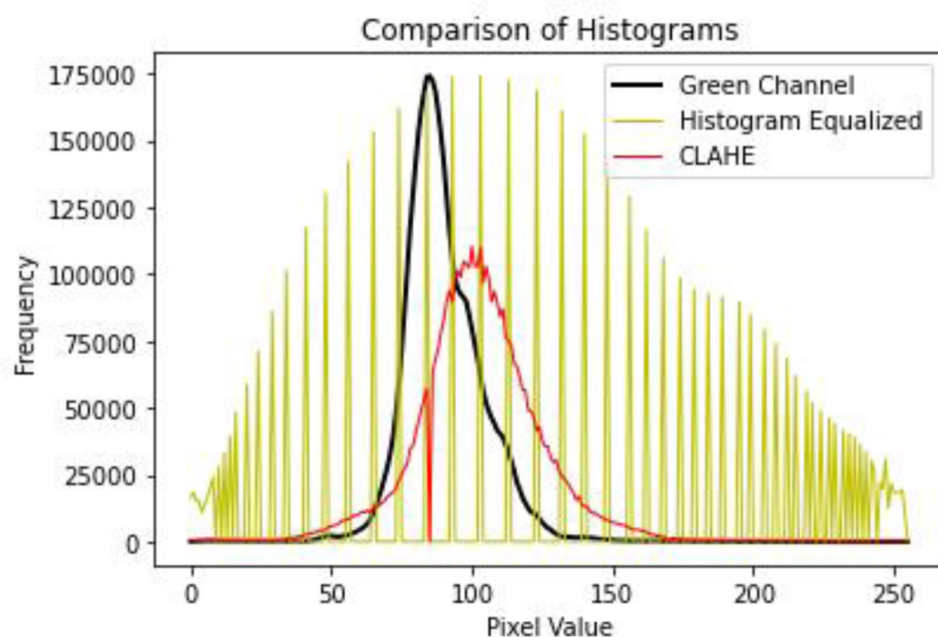


Fig. 4 Corresponding intensity histograms for images in the figure. CLAHE = contrast-limited adaptive histogram equalization.

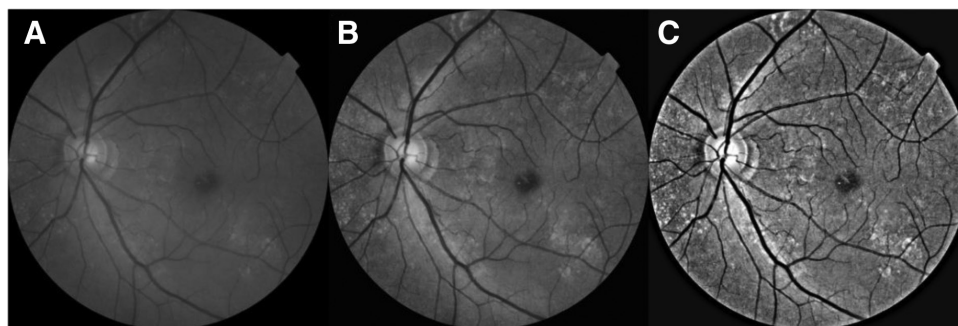


Fig. 5 Image sharpness comparison after application of CLAHE and unsharp masking. CLAHE = contrast-limited adaptive histogram equalization.

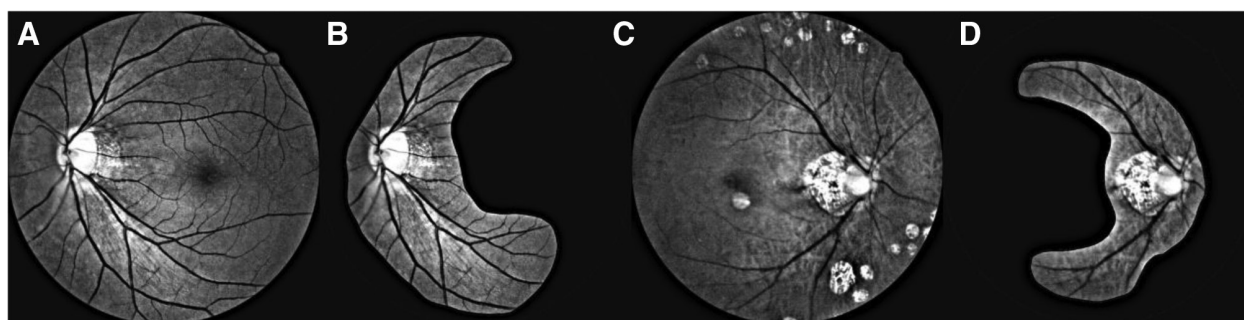


Fig. 6 Manually annotated ROI, including ONH and surrounding RNFL, with macular area excluded. ONH = optic nerve head; RNFL = retinal nerve fiber layer; ROI = regions of interest.

The segmentation model (DeepLabV3+ with MobileNet-V2 backbone) was trained on the ROI masks we defined. The resulting segmentation masks were used to generate weighted images, in which regions corresponding to the ROI retained their original pixel intensity and all other areas were down-weighted. This weighting strategy was designed to guide the classification model to focus on clinically relevant features and to reduce the effects of comorbidities and background noise.

We used a pretrained semantic segmentation model incorporating the DeepLabV3+ architecture to automatically segment the ROI. DeepLabV3+ was introduced by Google at the European Conference on Computer Vision in 2018. Its encoder-decoder structure and enhanced Xception model serve as the backbone network of encoders for image feature extraction. The encoder-decoder structure of the DeepLabV3+ architecture, illustrated in Fig. 7, contains several components designed for improved semantic segmentation, including the Atrous spatial pyramid pooling (ASPP) module. This module enables the model to capture features at multiple scales by applying dilated convolutions. As illustrated in Fig. 8, ASPP expands the receptive field coverage, improves feature extraction across resolutions, and reduces computational complexity. Depthwise separable convolution, illustrated in Fig. 9, is another crucial element of the DeepLabV3+ architecture. This feature combines depthwise and pointwise convolutions to reduce computational complexity while maintaining performance, a notable advantage in our study.

We used MobileNet-V2 instead of the Xception model as the backbone of the encoder network. Additionally, we applied Dice Loss²⁶ as the loss function and mean intersection-over-union (MIoU) as the training evaluation metric. MobileNet-V2 provides a lighter-weight network with fewer parameters and faster computation speed than Xception does. The Dice Loss

is defined in Equation (1), and the MIoU diagram is provided in Fig. 10.

$$\text{Dice Loss} = \frac{2 \sum_i^N p_i g_i}{\sum_i^N p_i^2 + \sum_i^N g_i^2} \quad (1)$$

where N is the number of the predicted binary segmentation pixels p_i and ground truth binary pixels g_i .

We applied a weighted image strategy to guide the model's focus on the ROI and reduce the effects of comorbidities. After ROI segmentation with the DeepLabV3+ model, the original pixel intensities within the ROI were retained, whereas those outside the ROI were down-weighted by 50%. This weighting approach preserved diagnostic information in relevant areas while suppressing noise from less informative regions. Fig. 11 illustrates this process, displaying the original image, segmentation output, and final weighted image.

2.4. Image classification with EfficientNetV2-M architecture and fine-tuning of process workflow

2.4.1. Application of EfficientNetV2-M architecture to fine-tune pretrained model

EfficientNetV2 was selected as the pretrained backbone for glaucoma prediction because of its favorable balance of accuracy and computational efficiency. EfficientNetV2 was introduced by Google at the 2021 International Conference on Machine Learning, and it achieved state-of-the-art performance on the ImageNet ILSVRC2012 and ImageNet21k benchmarks. Its key architectural components include the Fused-MBConv module; compound scaling strategies to expand depth, width, and resolution; and progressive learning techniques to accelerate training. These design features contribute to both improved classification accuracy and reduced computational cost.

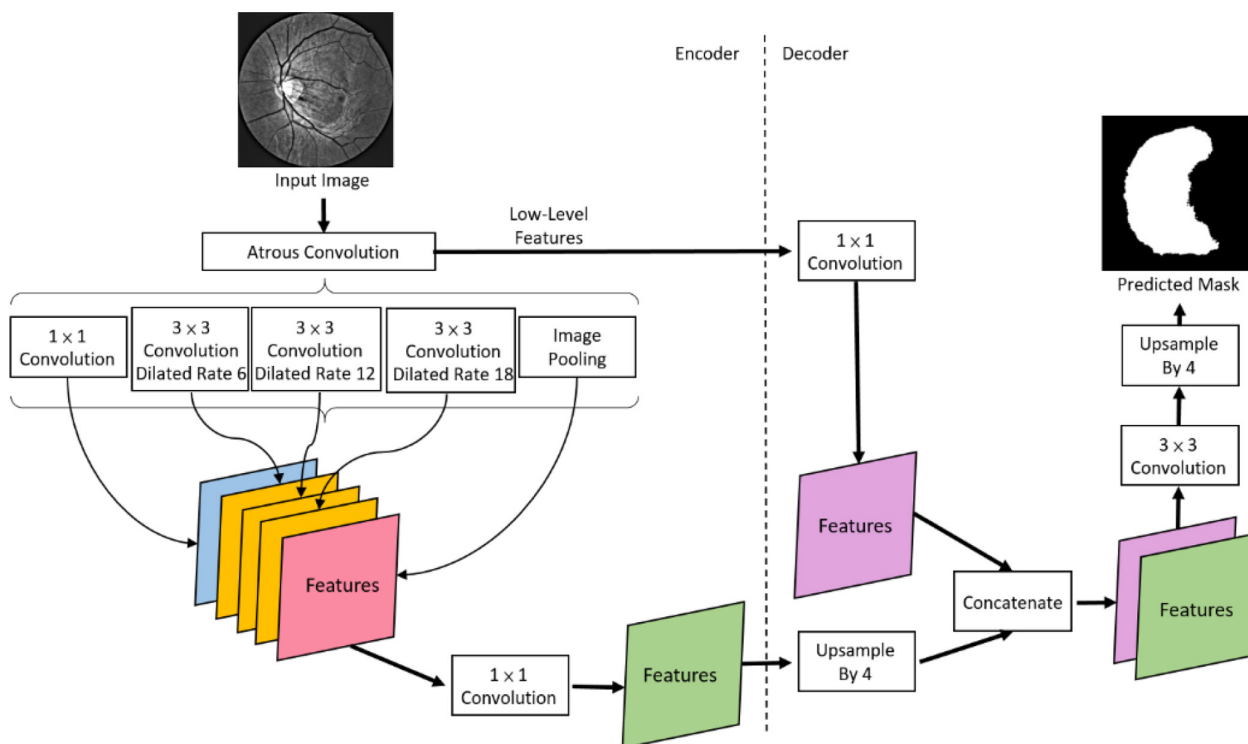


Fig. 7 Network architecture of DeepLabV3+ with MobileNet-V2 backbone for semantic segmentation.

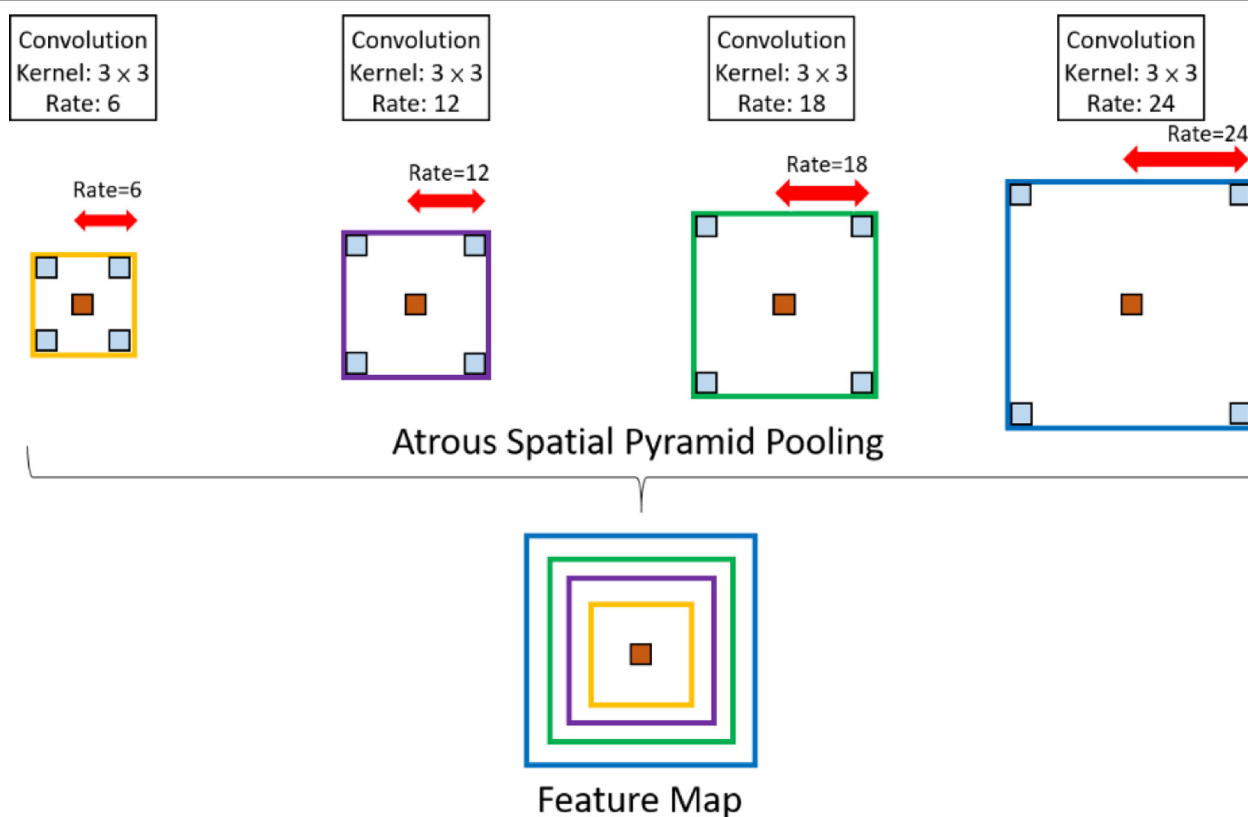


Fig. 8 Multiscale feature capture by ASPP module. ASPP = atrous spatial pyramid pooling.

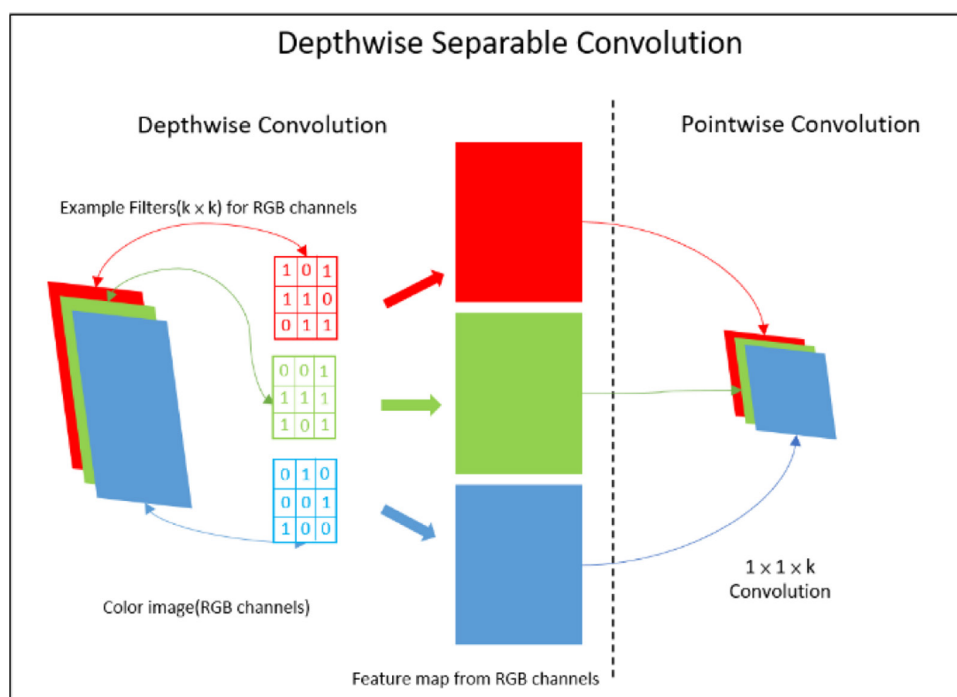


Fig. 9 Depthwise separable convolution structure combining depthwise and pointwise operations.

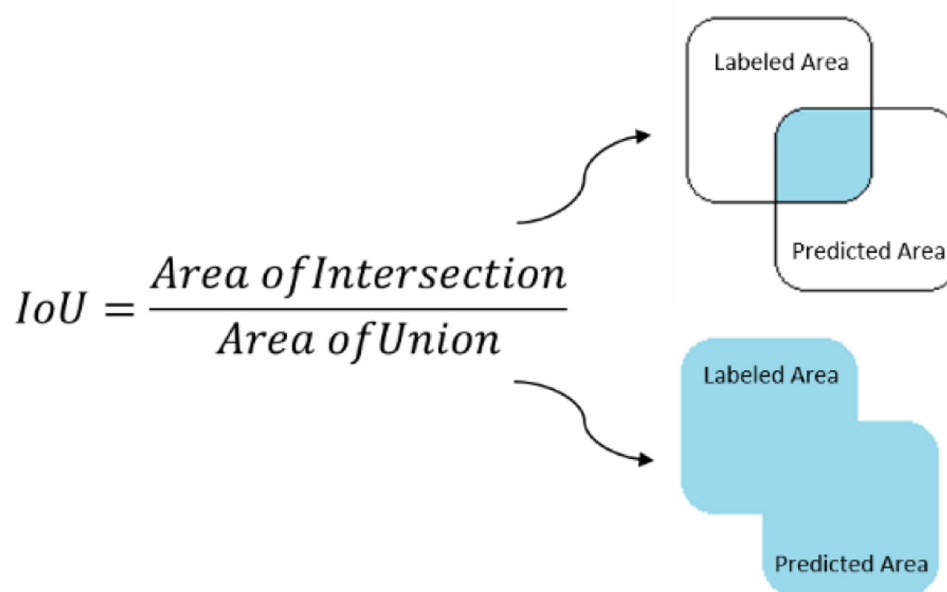


Fig. 10 MIoU calculation. MIoU = mean intersection-over-union.

We used a pretrained EfficientNetV2-M model as the glaucoma classification backbone to leverage its computational efficiency and suitability for deployment on resource-limited platforms. We retained the original architecture and replaced the final fully connected layer with a two-neuron output layer using a Softmax activation function for binary classification. The OpenCV Python library was used to resize images to the default resolution of 480×480 pixels, ensuring compatibility with the network and maintaining preprocessing efficiency, which improves the model's real-world applicability.

Focal Loss²⁷ was adopted as the loss function to address class imbalance during fine-tuning. Focal Loss extends standard cross-entropy (CE) loss by down-weighting easy examples to focus training on more difficult, misclassified cases, which is particularly beneficial in imbalanced medical datasets. For binary classification, Focal Loss is determined on the basis of the CE, expressed as follows:

$$CE(p, y) = CE(p_t) = -\log p_t$$

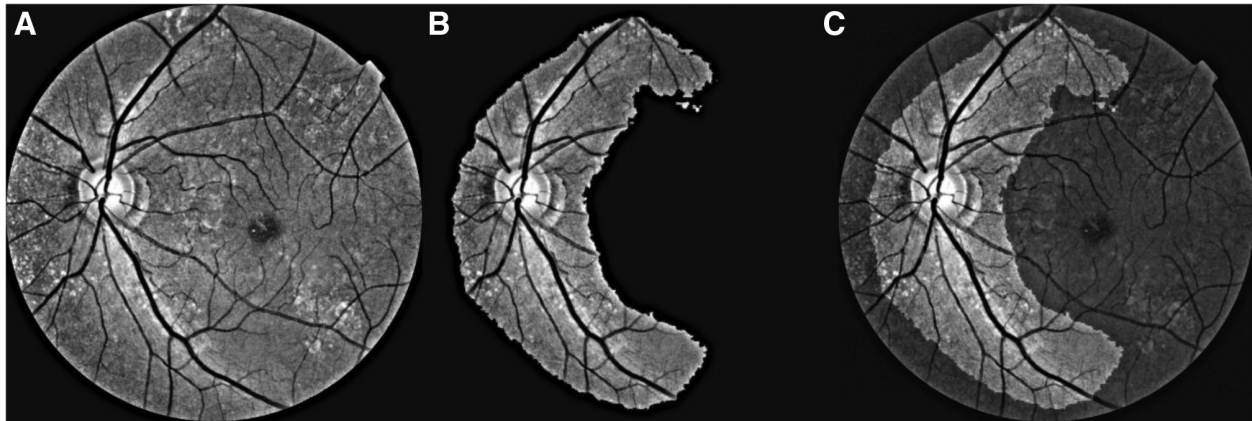


Fig. 11 Weighted image generation process: (A) original, (B) segmentation mask, (C) weighted image.

where $y \in \{\pm 1\}$ specifies the ground truth class and $p \in [0, 1]$ is the probability predicted by the model for the class labeled $y = 1$. p_t is defined as follows:

$$p_t = \begin{cases} -\log p, & y = 1 \\ -\log(1 - p), & \text{otherwise,} \end{cases}$$

Additionally, the α -balanced Focal Loss is defined as follows:

$$\alpha - \text{Balanced Focal Loss} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

where $\alpha \in [0, 1]$ for class 1 and $\gamma \geq 0$.

The learning rate decreases as the number of epochs increases, with a dynamic mechanism set to monitor the validation loss. The initial learning rate is 10^{-4} and remains fixed as long as the validation loss decreases. If the validation loss does not decrease for more than three epochs, the learning rate is reduced by 30%, with a minimum limit of 10^{-8} . We evaluated the model's predictive performance according to the area under the receiver operating characteristic curve (AUC ROC). The area under the curve (AUC) ranges from 0 to 1, with an AUC of 1 indicating a perfect classification model and an AUC of 0.5 indicating a model that randomly assigns categories to test data. The final model was selected on the basis of the AUC value of the validation set. Data augmentation was performed to improve model performance and prevent overfitting; we randomly rotated images by 20° , flipped them horizontally or vertically, and shifted them by 10% of the total width or height.

For validation, we applied TensorFlow, an open-source deep learning framework (Version 2.10.0, https://www.tensorflow.org/versions/r2.10/api_docs) with an NVIDIA GeForce RTX 3090 Ti graphics processing unit (NVIDIA Corporation, Santa Clara, CA). The accuracy, sensitivity, specificity, balanced accuracy, and AUC were calculated as follows:

Accuracy

$$= \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}}$$

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}$$

2.4.2. Youden index for optimal prediction thresholds and Grad-CAM for visualization of prediction results

During inference, the model outputs a probability score and applies a default classification threshold of 0.5 to distinguish between glaucomatous and normal cases. However, this default threshold may not yield optimal predictive performance, particularly in imbalanced or heterogeneous datasets. Given that the training, validation, and test sets in this study were derived through random sampling, we determined the optimal threshold on the basis of the validation set by using the Youden index, which balances sensitivity and specificity. The selected threshold was then applied to the test set to increase performance consistency. Given the clinical imperative of early glaucoma detection, a sensitivity of at least 80% was set as a requirement. Therefore, we only calculated the Youden index for thresholds that achieved a sensitivity $\geq 80\%$, ensuring the selected threshold met the minimum clinical criterion.

$$\text{Threshold}(T) = \left\{ \frac{i}{1000} : i = 0, 1, \dots, 999, 1000 \right\}$$

$$\text{Youden Index} = \max_{\tau \in T} (\text{Sensitivity}_\tau + \text{Specificity}_\tau - 1)$$

When this definition of the Youden index was applied, the recorded sensitivity exceeded the 80% threshold. The recorded threshold values were then used in sequence to calculate the optimal threshold for the best classification performance according to the Youden index.

Gradient-weighted class activation mapping (Grad-CAM)²⁸ was used to visualize the discriminative regions identified by the CNN classifier. Grad-CAM uses the gradients of the target class flowing into the final convolutional layer to compute the relative importance of each feature map. These importance weights are then combined with the feature maps to generate a coarse localization map, which is projected onto the input fundus image. The resulting heatmap highlights the primary regions contributing to the model's prediction, enabling users to interpret the classification decision.

2.5. Statistical analysis

CNN model performance was evaluated according to the MIoU, AUC, balanced accuracy, sensitivity, specificity, and accuracy

indices. The matplotlib package in Python was used to plot the ROC curve.

3. RESULTS

3.1. ROI segmentation performance

For ROI segmentation, a pretrained DeepLabV3+ model was fine-tuned on both original and preprocessed images. The model achieved an MIoU of 85.91% and 86.28% for original and preprocessed images, respectively, indicating that preprocessing improved performance. As indicated in Fig. 12, segmentation performance improved after image preprocessing was applied. Fig. 12A, B displays the original image and its corresponding predicted ROI. Fig. 12C, D displays the preprocessed input and the resulting segmented ROI. A comparison between Fig. 12B, D demonstrates the improved delineation of glaucoma-relevant structures following preprocessing. Additionally, the segmentation model was better able to suppress comorbid lesions and acquisition noise when trained with preprocessed images, as illustrated in Fig. 13.

3.2. Glaucoma detection performance

The classification performance of our model for five types of images contained in the test set of six cross-institutional datasets (TVGH, CHGH, DRISHTI-GS1, Refuge, LES-AV, and VEIRC) is presented in Table 2.

Table 2 summarizes the prediction performance of the TVGH-Original model, which served as the baseline. This model achieved a diagnostic accuracy of 90.03% on TVGH test images, with a balanced accuracy, sensitivity, specificity, and AUC of 88.90%, 85.19%, 92.61%, and 0.960, respectively. However, the diagnostic sensitivity of the TVGH-Original model decreased when tested within cross-institutional

datasets: CHGH (79.22%), DRISHTI-GS1 (31.43%), Refuge (67.50%), LES-AV (81.82%), and VEIRC (65.62%), with corresponding AUCs of 0.944, 0.844, 0.939, 0.909, and 0.882, respectively.

The cross-institutional datasets included images of individuals from diverse racial backgrounds. Thus, the green channel in the color fundus was used to fine-tune the model and increase the visibility of glaucoma lesions. The TVGH-Green Channel model exhibited notable improvements in sensitivity for the CHGH (85.71%), DRISHTI-GS1 (52.86%), and Refuge (82.50%) datasets.

Subsequently, the TVGH-preprocessing model, which enhanced contrast and sharpness in green channel images, was tested. The diagnostic sensitivity of this model on test images from TVGH, CHGH, DRISHTI-GS1, Refuge, LES-AV, and VEIRC reached 92.59%, 93.51%, 60.00%, 87.50%, 72.73%, and 59.36%, respectively. By comparison, the TVGH-Segmentation model demonstrated lower sensitivity than the TVGH-Preprocessing model did in the TVGH, DRISHTI-GS1, and LES-AV datasets, at 92.59%, 60.00%, and 72.73%, respectively. However, this lower sensitivity may have reduced interference from other retinal comorbidities. Finally, the TVGH-Weighted model achieved a sensitivity of over 70% and balanced accuracy of over 80% across all cross-institutional datasets, including datasets with comorbidities.

According to the experimental results, the cross-institutional prediction performance of the TVGH-Segmentation model was slightly lower than that of the TVGH-Preprocessing model, which was highly correlated with the segmentation model's performance. In our approach, the regions predicted by the segmentation model are overlaid onto the preprocessed images at the same ratio to eliminate comorbid lesions and noise and capture more imaging features. The brightness of the optic disc and nerve fibers

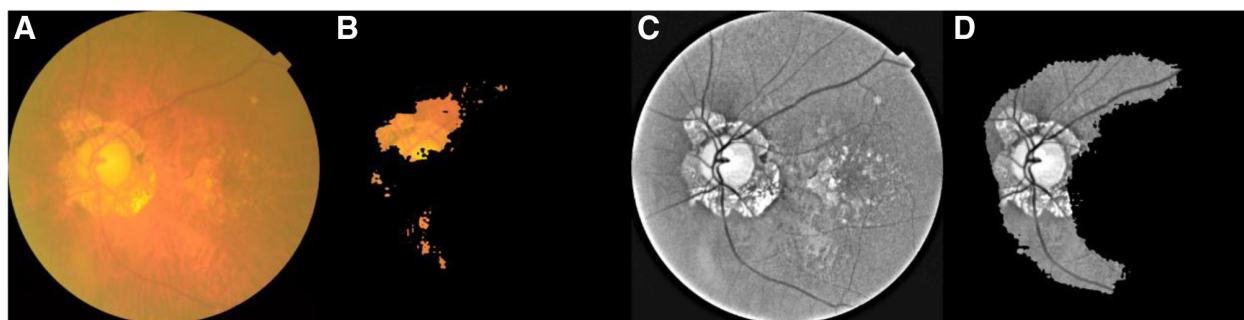


Fig. 12 Segmentation results before and after preprocessing: (A) and (B), results obtained with original images in external test sets; (C) and (D), results obtained with preprocessed images. Preprocessing improved segmentation performance, enabling more complete and accurate delineation of the optic disc and RNFL. RNFL = retinal nerve fiber layer.

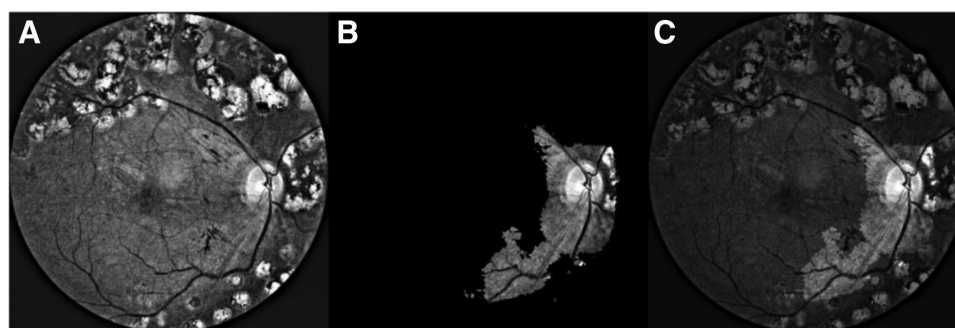


Fig. 13 Visualization of segmentation and weighting. (A) Original image, (B) segmentation prediction, and (C) resulting weighted image. The segmentation model successfully highlights glaucoma-relevant structures and suppresses regions affected by comorbidities or image noise.

Table 2**Comparison of model performance in internal and external datasets**

Model	Test datasets	Accuracy	Balanced accuracy	Sensitivity	Specificity	AUC (95% CI)
TVGH-Original	TVGH	90.03%	88.90%	85.19%	92.61%	0.960 (0.936-0.978)
	CHGH	87.90%	87.74%	79.22%	96.25%	0.944 (0.898-0.980)
	DRISHTI-GS1	51.49%	64.10%	31.43%	96.77%	0.844 (0.753-0.920)
	Refuge	94.75%	82.64%	67.50%	97.78%	0.939 (0.888-0.978)
	LES-AV	81.82%	81.82%	81.82%	81.82%	0.909 (0.750-1.000)
TVGH-Green Channel	VEIRC	91.44%	80.37%	65.62%	95.11%	0.882 (0.806-0.943)
	TVGH	88.75%	86.61%	79.63%	93.60%	0.968 (0.950-0.983)
	CHGH	90.45%	90.36%	85.71%	95.00%	0.959 (0.926-0.985)
	DRISHTI-GS1	65.35%	73.20%	52.86%	93.55%	0.891 (0.810-0.953)
	Refuge	94.50%	89.17%	82.50%	95.83%	0.941 (0.893-0.976)
TVGH-Preprocessing	LES-AV	77.27%	77.27%	63.64%	90.91%	0.860 (0.682-1.000)
	VEIRC	92.22%	78.13%	59.38%	96.89%	0.927 (0.870-0.969)
	TVGH	87.14%	88.85%	94.44%	83.25%	0.959 (0.937-0.977)
	CHGH	85.99%	86.08%	90.91%	81.25%	0.944 (0.904-0.978)
	DRISHTI-GS1	82.18%	82.65%	81.43%	83.87%	0.922 (0.855-0.972)
TVGH-Segmentation	Refuge	91.25%	85.14%	77.50%	92.78%	0.902 (0.817-0.971)
	LES-AV	81.82%	81.82%	81.82%	81.82%	0.917 (0.764-1.000)
	VEIRC	91.83%	76.57%	56.25%	96.89%	0.907 (0.840-0.960)
	TVGH	85.85%	87.43%	92.59%	82.27%	0.941 (0.911-0.967)
	CHGH	89.81%	89.88%	93.51%	86.25%	0.954 (0.917-0.984)
TVGH-Weighted	DRISHTI-GS1	70.30%	76.77%	60.00%	93.55%	0.901 (0.835-0.956)
	Refuge	79.25%	82.92%	87.50%	78.33%	0.904 (0.821-0.972)
	LES-AV	77.27%	77.27%	72.73%	81.82%	0.851 (0.639-1.000)
	VEIRC	85.99%	74.58%	59.36%	89.78%	0.807 (0.696-0.906)
	TVGH	87.14%	88.63%	93.52%	83.74%	0.963 (0.942-0.980)
	CHGH	84.71%	84.95%	97.40%	72.50%	0.961 (0.926-0.987)
	DRISHTI-GS1	79.21%	80.51%	77.14%	83.87%	0.889 (0.824-0.949)
	Refuge	84.00%	83.33%	82.50%	84.17%	0.913 (0.847-0.970)
	LES-AV	81.82%	81.82%	81.82%	81.82%	0.876 (0.673-1.000)
	VEIRC	88.33%	81.27%	71.88%	90.67%	0.887 (0.805-0.956)

The bold values indicate the key experimental results that highlight the most significant findings of this study.

AUC = area under the ROC curve; ROC = receiver operating characteristic.

remains consistent; that of other areas is reduced by 50%. The use of weighted imaging highlights glaucoma lesions, mitigates comorbidities, and preserves underlying image features. As indicated in Table 2, the TVGH-Preprocessing and TVGH-Weighted models achieved balanced accuracies close to or exceeding 80% during cross-institutional verification. However, no external datasets provided information on the proportion of glaucoma images with comorbidities. To further analyze how segmentation models can improve glaucoma detection performance on images with comorbidities, we invited glaucoma specialists to select 151 images from the

TVGH test set, with these comprising 28 glaucoma and 123 nonglaucoma images with comorbidities.

As illustrated in Table 3, all five models effectively detected glaucoma images without comorbidities. However, their predictive performance notably declined when the models were applied to images with comorbidities. Between the TVGH-Preprocessing model and TVGH-Weighted model—which exhibited the greatest cross-institutional verification performance—the TVGH-Weighted model demonstrated superior balanced accuracy for images with comorbidities, with an accuracy reaching 80.91%.

Table 3**Model AUC, sensitivity, specificity, and balanced accuracy in the dataset with comorbidities**

Model	TVGH test sets	Accuracy	Balanced accuracy	Sensitivity	Specificity	AUC
TVGH-Original	Glaucoma only	96.88%	96.88%	96.25%	97.50%	0.99
	Comorbidities	82.78%	71.50%	53.57%	89.43%	0.92
TVGH-Green Channel	Glaucoma only	93.75%	93.75%	91.25%	96.25%	0.99
	Comorbidities	83.44%	69.15%	46.43%	91.87%	0.93
TVGH-Preprocessing	Glaucoma only	97.50%	97.50%	100.00%	95.00%	0.99
	Comorbidities	76.16%	77.09%	78.57%	75.61%	0.90
TVGH-Segmentation	Glaucoma only	95.62%	95.62%	96.25%	95.00%	0.97
	Comorbidities	81.46%	78.96%	75.00%	82.93%	0.79
TVGH-Weighted	Glaucoma only	93.75%	93.75%	97.50%	90.00%	0.99
	Comorbidities	80.13%	80.91%	82.14%	79.67%	0.93

The bold values indicate the key experimental results that highlight the most significant findings of this study.

AUC = area under the ROC curve; ROC = receiver operating characteristic.

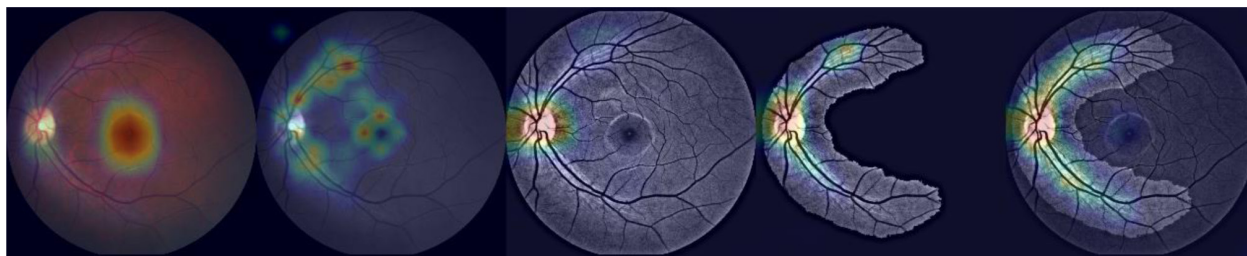


Fig. 14 Grad-CAM heatmaps illustrating regions critical for classification. Heatmaps were generated from test images processed through different stages (from right to left): original, green channel, preprocessed, segmented, and weighted images. Grad-CAM highlights the predominant diagnostic regions contributing to the model's decision, particularly around the ONH. Grad-CAM = gradient-weighted class activation mapping; ONH = optic nerve head.

3.3. Grad-CAM for visualization of model prediction results

As demonstrated in Fig. 14, Grad-CAM was applied to highlight model features critical for classification, with these depicted using heat maps. These features were primarily concentrated in the ONH and peripapillary nerve fibers.

4. DISCUSSION

We developed a glaucoma detection model to address the challenges posed by cross-institutional image variability and interference from retinal comorbidities. Our weighted segmentation approach demonstrated high generalizability across multiple external datasets and was validated using a limited number of training images from a single institute. Our approach achieves generalizability through the use of image preprocessing and segmentation without the need for extensive fine-tuning involving large quantities of images from various external sources. These image processing methods are rooted in our domain knowledge of the diagnostic cues and pathogenesis of glaucoma. Our strategy has notable advantages and may provide medical institutions—including institutions with limited resources or image data for fine-tuning their models—with a reliable AI-powered diagnostic tool for glaucoma.

Our robust glaucoma feature extraction process was developed using color fundus images, and our method achieved comparable diagnostic performance to strategies involving OCT images.¹⁵ Fundus images, which can be obtained with low-cost, portable equipment, have considerable potential for glaucoma screening. Additionally, our findings suggest that in scenarios involving data imbalance and limited external validation samples, accuracy and AUC metrics may insufficiently reflect model performance. For example, as indicated in Table 2, the sensitivity of the TVGH-Green Channel and TVGH-Preprocessing models on the VEIRC dataset was 59.38% and 56.25%, respectively. However, their accuracies reached 92.22% and 91.83%, with corresponding AUC values of 0.927 and 0.907. Therefore, we suggest that balanced accuracy may be a more robust evaluation metric for model performance.

As expected, the performance of the TVGH-Original model markedly declined when applied to external datasets. We therefore developed the TVGH-Green Channel model, which leverages the image's green channel for enhanced visualization of the RNFL, a crucial characteristic in glaucoma evaluation. This model demonstrated improved balanced accuracy in the CHGH, DRISHTI-GS1, and Refuge datasets. Within the DRISHTI-GS1 dataset, both the model's sensitivity and balanced accuracy considerably increased. These experimental results indicate that the features the model learns from the green channel images may not improve overall prediction

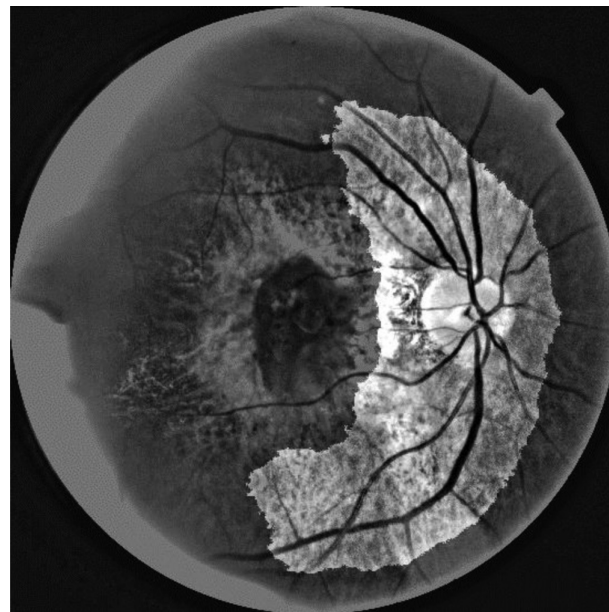


Fig. 15 Segmentation results demonstrating the model's ability to exclude comorbid lesions and noise.

performance but can reduce cross-institutional variabilities in color fundus images.

The TVGH-preprocessing model combines green channel images with image processing methods, offering enhanced contrast and sharpness to improve visualization of glaucoma lesions. This model demonstrated a balanced accuracy above 80% in most test data, that is, all but the VEIRC dataset. Matching of green channel contrast and sharpness thus improved the model's generalization during cross-institutional verification.

As illustrated in Fig. 15, the segmentation model effectively excluded comorbidity lesions and noise during testing. This model accurately segmented the optic disc and optic nerve fibers in images centered on either the disc or macula, as displayed in Fig. 16. For the TVGH-Segmentation model, the regions predicted by the segmentation model were used as the training set. The balanced accuracy of its classification within the TVGH test set and CHGH and Refuge datasets reached 87.43%, 89.88%, and 82.92%, respectively. In the DRISHTI-GS1, LES-AV, and VEIRC datasets, it achieved balanced accuracy rates of 76.77%, 77.27%, and 74.58%, respectively. These results indicate that the TVGH-Segmentation model needs to only focus on the areas marked by the segmentation model to achieve a balanced accuracy of at least 74.58% in cross-institutional verification.

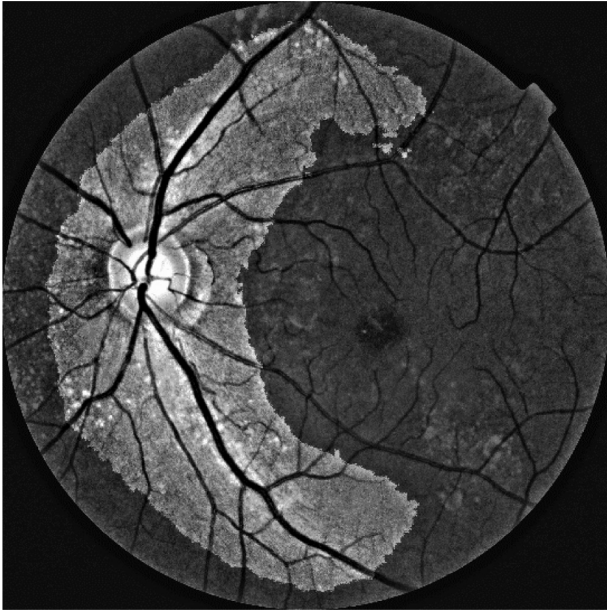


Fig. 16 Segmentation robustness across image centers.

Although the TVGH-Segmentation model effectively reduces the confounding effects of comorbid lesions, it may inadvertently diminish the perception of glaucoma lesions beyond the optic disc and nerve fibers. Thus, in the TVGH-Weighted model, the brightness of the automatically segmented area is maintained, whereas the brightness of all other regions is reduced. This increases the weight of glaucoma lesions while minimizing the effects of comorbid lesions and noise. We trained the TVGH-Weighted model with a higher weight assigned to the area surrounding glaucoma lesions. The experimental results presented in Tables 2 and 3 indicate that the TVGH-Weighted model achieved a balanced accuracy of over 80% for color fundus images in cross-institutional validation datasets and datasets with comorbidities. Our findings regarding the TVGH-Weighted model demonstrate the potential of AI-assisted diagnosis in clinical applications, indicating it exhibits promise for improved glaucoma detection in real-world scenarios.

Notably, Al-Bander et al¹¹ and Pachade et al¹² have used deep learning approaches to segment optic cups and discs from fundus images. Although the CDR is crucial for glaucoma evaluation, it is influenced by disc size and cannot serve as the sole parameter for diagnosis. RNFL patterns and changes in color fundus images also provide valuable diagnostic information. Our image preprocessing and segmentation model is designed to efficiently extract information from the optic disc and RNFL. Similarly, Ko et al¹³ used a pretraining model for glaucoma detection and fine-tuned the model weights by integrating external data with the existing training dataset. This approach produced a glaucoma detection model applicable across institutions. However, using this type of mixed-data approach raises concerns regarding information security and does not address the confounding effects of comorbidities. Our study presents a robust feature extraction method that does not require external data to fine-tune weights, featuring a segmentation approach suitable for images with comorbidities. Qian et al¹⁴ adopted a different segmentation approach in which the radius of the optic disc is computed on the basis of the U-Net architecture, with a rectangular area three times that of the radius serving as the ROI. This method was validated on two external datasets from Chinese institutions, yielding AUCs of 0.952 and

0.966, respectively. The higher ROC observed in Qian et al¹⁴ study may be related to the requirement they imposed of images passing quality checks before classification. However, whether their approach is affected by comorbidities involving this area remains unclear. Nevertheless, the aforementioned studies validate the effectiveness of the segmentation approach.

The TVGH-Weighted model was validated on an external dataset from Taiwan (CHGH), achieving an AUC of 0.961. It was also tested on multinational external datasets, including DRISHTI-GS1, Refuge, LES-AV, and VEIRC, yielding AUCs of 0.889, 0.913, 0.876, and 0.887, respectively. Verification of the TVGH-Weighted model on a dataset with comorbidities produced an AUC of 0.93. These findings indicate that our proposed TVGH-Weighted model is generalizable to cross-institutional retinal images from individuals of various ethnicities and those with comorbidities, which underscores the feasibility of applying AI-assisted technology in clinical diagnosis. Our method addresses cross-institutional efficiency challenges in AI-assisted diagnosis and enhances green channel image features. To mitigate the effects of comorbidities and noise on model performance, the segmentation model identifies critical areas for glaucoma diagnosis, including the optic disc and the region around major vascular trunks representing the RNFL bimodal peaks. This data-driven approach mitigates the confounding effects of comorbid lesions and noise during ROI selection, enabling the model to devote comprehensive focus to vital information regarding the optic disc and its surroundings.

Image quality represents a potential limitation that may hinder the clinical application of our approach. ROI segmentation must be implemented to reduce interference from comorbidities and noise, but low-quality images may hinder the segmentation algorithm's ability to accurately extract the ROI for classification, which can adversely affect diagnostic performance. Although we verified our model using several open-access external datasets to demonstrate its generalizability, large-scale verification must be implemented to confirm its real-world performance.

Our segmentation model is designed to suppress the influence of comorbid lesions by focusing on the ONH and peripapillary RNFL. However, the external datasets we used for validation do not contain structured labels for comorbid retinal conditions. Consequently, our direct evaluation of classification performance in the presence of comorbidities was limited to 151 internally annotated images, constraining the full validation of our approach in heterogeneous real-world populations. Therefore, future work should include prospectively labeled multi-institutional cohorts or simulation-based evaluations. This would enable systematic confirmation of model performance in the presence of diverse retinal comorbidities.

In this study, our exclusion of overexposed discs required manual visual assessment; objective brightness metrics can be incorporated in future studies to improve reproducibility. Our segmentation masks, which included major vessel trunks to encompass RNFL bimodal peaks, improved the balanced accuracy in comorbidity-laden images compared with the original model (80.91% vs 71.50%, Table 3); however, we did not perform an explicit ablation study to isolate the effects of vessel inclusion on model performance in comorbidity-laden images. In future studies, this design choice should be systematically evaluated to further optimize diagnostic accuracy.

In conclusion, this study presents a robust and cost-effective AI-assisted framework for glaucoma detection in color fundus images. The proposed method combines domain-driven image preprocessing with semantic segmentation, effectively mitigating the effects of retinal comorbidities and cross-institutional image variability. Our model achieved at least 80% balanced accuracy in experimental testing across five external datasets,

which included patients from diverse ethnic backgrounds and imaging conditions. Notably, segmentation-based weighting markedly increased diagnostic performance in challenging cases with comorbidities. These results highlight the potential of lightweight, generalizable AI models to enable accurate glaucoma screening in fundus images alone, which are widely accessible. This approach has strong potential for deployment in real-world clinical settings, particularly in resource-limited environments.

Declaration of generative AI in scientific writing

The authors used ChatGPT to prepare the manuscript and subsequently reviewed and edited the content as necessary. The authors take full responsibility for the content of the publication.

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